Functions, Rules, Domain & Range

WARM UPS

TEACHER COPY | Display student pages; keep solutions private

1 | Solve

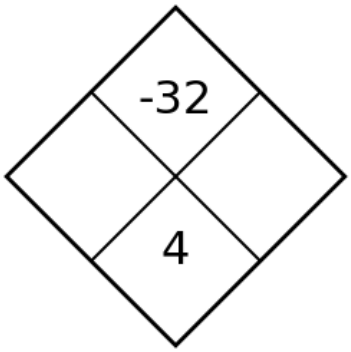
Solve for x. Show each step, then check your answer.

$$7x - 4 + 2x = 23$$

> **ANSWER / TEACHING NOTE**
> 7x - 4 + 2x = 23 → 9x - 4 = 23 → 9x = 27 → x = 3.

2 | Diamond

Complete the diamond.



Functions, Rules, Domain & Range

WARM UPS

3 | Spiral Review

Evaluate $2a^2 - 3$ when $a = -2$. Show substitution.

> **ANSWER / TEACHING NOTE**
> 2(-2)^2 - 3 = 2(4) - 3 = 5.

4 | Spiral Review

A pattern has outputs 5, 8, 11, 14 for inputs 0, 1, 2, 3. What is the change in output each step?

> **ANSWER / TEACHING NOTE**
> The output increases by 3 for each +1 input.

Functions, Rules, Domain & Range

LEARNING FOCUS & NOTES

Learning focus | LG 02-ican-1 / LG 02-ican-2 / LG 02-ican-3

- I can use an input-output rule to find outputs.
- I can explain why a relation is or is not a function.
- I can list the domain and range of a set of ordered pairs.

Vocabulary | Highlight the important words

Relation
A collection of input-output pairs; it may or may not be a function.

Function
A relation that assigns exactly one output to each input.

Input / output
An input is the starting value (often x); an output is the resulting value (often y).

Domain
The set of all distinct input values used by a relation.

Range
The set of all distinct output values produced by a relation.

Function rule
An equation or instruction that tells how to get an output from an input.

Meaning & Method | Rewritten review notes

1. Read an ordered pair as (input, output). A repeated input is allowed only if it gives the SAME output.

2. To test a function, inspect whether any input has two different outputs. On a graph, a vertical line should not hit more than one point.

3. For a finite relation, list each domain and range value only once; order from least to greatest.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

Functions, Rules, Domain & Range

EXAMPLES & YOU TRY ITS

Example

Is this relation a function? Explain why, then identify its domain and range.

Input x	Output y
-1	4
0	3
1	2
2	1

Decide if this relation is a function. List the domain and range.

> **ANSWER / TEACHING NOTE**
> Yes. Every input has only one output. Domain = {-1, 0, 1, 2}; range = {1, 2, 3, 4}.

You Try It

Determine whether this table represents a function. State the domain and range, and justify your answer.

Input x	Output y
-2	1
0	3
0	5
3	7

Function? Explain. Domain? Range?

> **Function? Explain. Domain? Range?**
> Not a function: the input 0 has outputs 3 and 5. Domain = {-2, 0, 3}; range = {1, 3, 5, 7}.

Why does the repeated input in this table matter?

> **TEACHING NOTE**
> Input 0 has two distinct outputs, 3 and 5, so the table is not a function. A duplicate pair with the same output would be allowed.

Functions, Rules, Domain & Range

LAUNCH

Function detective: four cases

Work with a partner. For each case, decide whether the relation is a function, then record its domain and range.

Case A

x	y
1	2
2	4
3	6

Case B

x	y
1	5
1	7
2	9

Case C

x	y
0	4
1	4
2	4

Case D

x	y
1	2
1	2

1 | Compare Cases B and D. Both repeat input 1. Why does only one fail the function test?

> **ANSWER / TEACHING NOTE**
> A, C, and D are functions; B is not. In B, input 1 has two different outputs (5 and 7). In D, input 1 repeats with the same output (2), so the relation remains a function. D has domain {1} and range {2}. A has domain {1,2,3} and range {2,4,6}; C has domain {0,1,2} and range {4}.

2 | Choose one function and predict another input-output pair from a pattern you observe. Is your new pair certain or a reasonable prediction? Explain.

> **ANSWER / TEACHING NOTE**
> Example for A: (4,8) follows a possible pattern, but the given table does not prove it must continue.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Functions, Rules, Domain & Range

REVIEW & PREVIEW

1 A function machine follows $y = 2x + 5$. Complete the outputs for inputs 0, 2, 4, and 6.

> **ANSWER / TEACHING NOTE**
> Outputs: 5, 9, 13, 17.

2 ★ For the relation {(-1,0), (0,2), (-1,3), (2,8)}, decide whether it is a function. Explain and list domain/ range.

> **ANSWER / TEACHING NOTE**
> Not a function: -1 maps to 0 and 3. Domain {-1, 0, 2}; range {0, 2, 3, 8}.

3 Write two different ordered pairs with the same OUTPUT but different INPUTS. Does this break the function rule?

> **ANSWER / TEACHING NOTE**
> Example: (1,4) and (2,4). Yes, this can be a function; outputs may repeat.

Functions, Rules, Domain & Range

REVIEW & PREVIEW

4 ★ A game costs $5 to enter, plus $3 per round. Use inputs 0, 1, 2, 3 rounds to list the outputs and the domain/range.

> **ANSWER / TEACHING NOTE**
> Outputs: 5, 8, 11, 14. Domain {0, 1, 2, 3}; range {5, 8, 11, 14}.

5 Spiral: Solve $2(x+5)=24$. Show your steps.

> **ANSWER / TEACHING NOTE**
> 2(x+5)=24 → x+5=12 → x=7.

6 Preview: Table outputs are 9, 6, 3, 0 for inputs 1, 2, 3, 4. What stays constant as the input increases by 1?

> **ANSWER / TEACHING NOTE**
> Constant difference is -3 for each increase of 1 input.

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.